

Scientific Measurement – The Foundation of Reliable Science

Chemistry is a quantitative science. It is not enough to say that a reaction produces "a little bit" of gas or that a solution is "somewhat hot." Chemists measure everything — mass, volume, temperature, concentration, time. The numbers they obtain mean nothing unless the measurements are accurate, precise, and reported with the correct units and proper significant figures. Yet many students rush through measurements without understanding why careful technique matters.

Chemistry is a science of precise measurement. From the pharmacist measuring milligrams of a life-saving drug to the environmental chemist detecting parts per billion of a pollutant, from the industrial engineer controlling reactor temperatures to the forensic scientist analyzing trace evidence, measurement governs every aspect of chemical practice. Understanding scientific measurement means understanding how to obtain trustworthy data — and how to tell good data from bad.

This reading material explores the differences between mass, volume, and density, the concepts of accuracy and precision, sources of experimental error, the use of scientific notation, and the rules for determining significant figures in calculations.

Part 1: What Is Scientific Measurement and Why Does It Matter?

Definition of Scientific Measurement

Scientific measurement is the process of obtaining quantitative data (numbers with units) about physical properties such as mass, volume, length, temperature, and time. In chemistry, measurements must be reproducible, comparable, and meaningful.

The Central Idea: Measurements Must Be Standardized

If every country used different units, scientists could not share data. If every balance gave different readings, experiments could not be repeated. Standard units (metric system, SI units) and proper techniques ensure that a measurement made in one laboratory means the same thing in another laboratory.

Why Measurement Matters in Chemistry

Aspect	Explanation
Reproducibility	Other scientists must be able to repeat experiments and obtain similar results
Communication	Standard units allow chemists worldwide to share data without confusion
Safety	Incorrect measurements (too much reagent, wrong temperature) can cause explosions or toxic reactions
Quality control	Manufacturing, medicine, and food production depend on precise measurements
Law and regulation	Environmental limits (e.g., pollutant concentrations) require accurate measurement

Part 2: Mass, Volume, and Density of Solids

Mass

Mass is the amount of matter in an object. It does not change with location or gravity. A rock has the same mass on Earth as on the Moon; only its weight changes.

Feature	Explanation
Definition	The quantity of matter in an object
SI unit	Kilogram (kg)
Common laboratory unit	Gram (g) (1 g = 0.001 kg)
Instrument	Balance (triple beam, electronic top-loading, analytical)
Examples	1 mL of water has a mass of approximately 1 gram
What mass does NOT depend on	Gravity, location, shape

Volume

Volume is the amount of space an object occupies. For liquids, volume is measured using graduated cylinders, pipettes, or burettes. For solids, volume can be measured directly (for regular shapes using formulas) or by displacement (for irregular shapes).

Feature	Explanation
Definition	The amount of space an object occupies
SI unit	Cubic meter (m ³)
Common laboratory units	Liter (L), milliliter (mL), cubic centimeter (cm ³)
Relationship	1 mL = 1 cm ³ (exactly)
Instrument (liquids)	Graduated cylinder, pipette, burette, volumetric flask
Instrument (regular solids)	Ruler (calculate from length × width × height)
Instrument (irregular solids)	Water displacement in a graduated cylinder

Density

Density is the mass of a substance per unit volume. It is an intensive property — it does not depend on how much substance you have. Density can be used to identify unknown substances.

Feature	Explanation
Definition	Mass per unit volume
Formula	Density = Mass / Volume ($D = m/V$)
Units	g/mL (or g/cm ³) for solids and liquids; g/L for gases
What it tells you	How tightly packed the matter is
Intensive vs. extensive	Density is intensive (does not change with sample size); mass and volume are extensive

Calculating Density of a Solid

Step	Procedure
1	Measure the mass of the solid using a balance (record in grams)
2	Measure the volume of the solid (using ruler for regular shape or displacement for irregular shape)
3	Divide mass by volume ($D = m/V$)
4	Report density with correct units (g/cm ³ or g/mL)

Example Calculation

Given:

Mass of a metal cube = 54.0 g

Length = 3.0 cm

Width = 3.0 cm

Height = 3.0 cm

Step 1: Calculate the volume of the cube.

Volume = length $\times$ width $\times$ height

Volume = 3.0 cm $\times$ 3.0 cm $\times$ 3.0 cm = 27.0 cm³

Step 2: Write the formula for density.

Density = Mass / Volume

Step 3: Substitute the values into the formula.

D = 54.0 g / 27.0 cm³

Step 4: Perform the division.

D = 2.0 g/cm³

Step 5: Compare to known values.

This density matches the density of aluminum (2.70 g/cm³ within experimental error).

Comparison Table: Mass, Volume, and Density

Property	Definition	Common Units	Instrument	Intensive or Extensive?
Mass	Amount of matter	g, kg	Balance	Extensive (depends on sample size)

Volume	Space occupied	mL, L, cm^3	Graduated cylinder, ruler	Extensive (depends on sample size)
Density	Mass per unit volume	g/mL , g/cm^3	Calculated (mass $\div$ volume)	Intensive (does not depend on sample size)

Part 3: Investigating the Density of Solids – A Scientific Report

When investigating the density of commonly used solids, a **scientific report** must be written. The standard format includes the following sections.

Scientific Report Format for Density Investigation

Section	Description	What to Include
Title	Clear statement of the investigation	Example: "Determining the Density of Regular and Irregular Solids"
Aim (or Purpose)	What you want to find out	Example: "To calculate the density of three solid samples using mass and volume measurements"
Hypothesis	Prediction based on prior knowledge	Example: "Each solid will have a unique density that can be used to identify the material"
Materials	List of equipment used	Balance, graduated cylinder, ruler, water, irregular solid (e.g., rock), regular solid (e.g., metal cube)

Section	Description	What to Include
Method (Procedure)	Step-by-step instructions written in past tense	1. The mass of the solid was measured using a balance. 2. The volume was measured by displacement. 3. Density was calculated using $D = m/V$.
Results (Data)	Organized table of measurements	Table showing mass, volume, and calculated density for each solid
Calculations	Show the formula and worked examples	$D = m/V = 54.0 \text{ g} / 27.0 \text{ cm}^3 = 2.0 \text{ g/cm}^3$
Discussion	Interpret results, identify errors, suggest improvements	Compare experimental density to known values; explain sources of error
Conclusion	Summary of findings	"The density of aluminum was found to be 2.0 g/cm^3 , which matches the accepted value of 2.70 g/cm^3 within experimental error."

Sample Data Table for Density Investigation

Solid Sample	Mass (g)	Volume (cm ³)	Density (g/cm ³)
Aluminum cube	54.0	27.0	2.0
Iron nail	15.8	2.0	7.9
Wood block	12.0	24.0	0.50

Part 4: Accuracy and Precision in Experimental Results

Accuracy and precision are two different ways of assessing the quality of measurements. Both are essential for reliable science.

Definitions

Term	Definition	Analogy
Accuracy	How close a measurement is to the true or accepted value	Hitting the bullseye on a dartboard
Precision	How close repeated measurements are to each other (reproducibility)	Hitting the same spot on the dartboard repeatedly, even if it is not the bullseye

Why Both Accuracy and Precision Matter

Situation	Problem
Accurate but not precise	The average is correct, but individual measurements vary too much; you cannot trust any single measurement
Precise but not accurate	The measurements are consistent, but all are wrong due to systematic error (e.g., uncalibrated balance)
Both accurate and precise	Ideal situation; measurements are correct AND reproducible
Neither	Data is useless; experiment must be redesigned

Example in Chemistry

Scenario	Accuracy?	Precision?	Problem
A balance reads 10.00 g for a 10.00 g standard mass	High	High	No problem
A balance reads 9.95 g, 9.96 g, 9.94 g for a 10.00 g standard	Low (off by 0.05 g)	High (values close together)	Systematic error (balance needs calibration)
A balance reads 10.02 g, 9.85 g, 10.15 g for a 10.00 g standard	Medium (average is close)	Low (values spread out)	Random error (poor technique)

Part 5: Sources of Error and Improvements to Enhance Reliability

No experiment is perfect. Identifying sources of error and proposing improvements is a key skill in chemistry.

Types of Error

Type of Error	Definition	Example
Systematic error	Consistent error in one direction (always too high or too low); caused by equipment or method	Uncalibrated balance always reads 0.05 g too high; thermometer reads 2°C above true temperature

Random error	Unpredictable variation in either direction; caused by human technique or environmental fluctuations	Slight differences in reading a meniscus; temperature changes in the lab
Gross error	Large mistake (not considered a true error)	Spilling a sample; misreading a number; using the wrong unit

Identifying Sources of Error in Density Experiments

Source of Error	Type	Explanation	Improvement
Balance not zeroed (tared)	Systematic	All mass readings are off by the same amount	Calibrate balance before use; check zero
Parallax error reading the meniscus	Random	Reading volume from an angle gives different values	Read at eye level; use the bottom of the meniscus
Air bubbles on irregular solid during displacement	Random	Bubbles take up volume, making solid appear larger	Gently tap the solid to remove bubbles
Using an inaccurate graduated cylinder	Systematic	A 100 mL cylinder is less precise than a 10 mL cylinder	Choose appropriate equipment
Solid not completely submerged	Systematic	Volume measured is too small	Ensure solid is fully under water
Water evaporates during measurement	Random	Volume decreases slightly over time	Work quickly; cover the cylinder

How to Propose Improvements

Step	What to Do	Example
1	Identify the specific error	"The volume measurements varied because the meniscus was read from different angles."
2	Explain why it affects reliability	"This introduced random error, reducing precision."
3	Propose a specific improvement	"All readings should be taken at eye level using the bottom of the meniscus."
4	Suggest a control if applicable	"Three trials should be conducted and averaged."

Enhancing Measurement Reliability

Strategy	How It Helps
Repeat measurements (multiple trials)	Reduces random error; allows calculation of average
Use calibrated equipment	Eliminates systematic error from uncalibrated instruments
Use appropriate equipment	A 10 mL graduated cylinder is more precise for 5 mL than a 100 mL cylinder
Train consistent technique	Reduces human variation
Control environmental conditions	Temperature and humidity can affect measurements (e.g., volume of liquids)

Part 6: Scientific Notation and Significant Figures

Scientific notation and significant figures ensure that measurements are reported honestly and that calculations reflect the precision of the original data.

Scientific Notation

Scientific notation expresses very large or very small numbers as a product of a number between 1 and 10 and a power of ten.

Rule	Example
Move the decimal so only one non-zero digit remains to the left	$0.000450 \rightarrow 4.50$
Count how many places the decimal moved	Moved 4 places to the right
Write as (number) $\times 10^{\text{(exponent)}}$	4.50×10^{-4}
Positive exponent for large numbers (moved left)	$5,600,000 = 5.6 \times 10^6$
Negative exponent for small numbers (moved right)	$0.0000032 = 3.2 \times 10^{-6}$

Converting Numbers to Scientific Notation

Original Number	Move Decimal	Scientific Notation
602,000,000,000,000,000,000	23 places left	6.02×10^{23} (Avogadro's number)
0.00000000000000000000000167	24 places right	1.67×10^{-24} (mass of a proton in grams)
0.00560	3 places right	5.60×10^{-3}

Original Number	Move Decimal	Scientific Notation
450,000	5 places left	4.5×10^5

Significant Figures (Significant Digits)

Significant figures are the digits in a measurement that are known with certainty plus one estimated digit. They indicate the precision of the measurement.

Rules for Determining Significant Figures

Rule	Explanation	Examples
1	All non-zero digits are significant	123 has 3 sig figs; 4.5 has 2 sig figs
2	Zeros between non-zero digits are significant	101 has 3 sig figs; 1.002 has 4 sig figs
3	Leading zeros (before first non-zero) are NOT significant	0.0012 has 2 sig figs; 0.000450 has 3 sig figs
4	Trailing zeros after a decimal point ARE significant	12.00 has 4 sig figs; 0.0450 has 3 sig figs
5	Trailing zeros without a decimal point are NOT significant (ambiguous)	1200 has 2 sig figs (unless written as 1200. with decimal)

Special Cases for Rule 5 (Trailing Zeros)

Number	Significant Figures	Reason
1200	2 (1.2×10^3)	No decimal; zeros are not significant
1200.	4 (1.200×10^3)	Decimal present; zeros are significant

Number	Significant Figures	Reason
0.001200	4	Leading zeros not significant; trailing zeros after decimal ARE significant

Significant Figures in Calculations

Operation	Rule	Example
Multiplication and Division	Result has the same number of significant figures as the measurement with the FEWEST sig figs	$2.50 \text{ cm} \times 3.0 \text{ cm} = 7.5 \text{ cm}^2$ (2.50 has 3 sig figs; 3.0 has 2 sig figs → answer has 2 sig figs)
Addition and Subtraction	Result has the same number of DECIMAL PLACES as the measurement with the FEWEST decimal places	$12.11 \text{ g} + 0.3 \text{ g} = 12.4 \text{ g}$ (12.11 has 2 decimal places; 0.3 has 1 decimal place → answer has 1 decimal place)
Exact numbers (counted, definitions)	Do not affect sig figs (infinite sig figs)	$12 \text{ eggs (exact)} \times 2.00 \text{ g} = 24.0 \text{ g}$ (keeps sig figs of measured number)

Rounding Rules for Significant Figures

Rule	Example
If the digit after the last significant digit is less than 5, round down	$3.342 \rightarrow 3.34$ (3 sig figs)
If the digit after the last significant digit is 5 or greater, round up	$3.347 \rightarrow 3.35$ (3 sig figs)
For 5 exactly, round to nearest even digit (to avoid bias)	$3.345 \rightarrow 3.34$ (not 3.35)

Worked Example: Density Calculation with Sig Figs

Step	Value	Sig Figs
Mass measured	54.00 g	4 sig figs
Volume measured	27.0 cm ³	3 sig figs
Density calculation	$54.00 \text{ g} \div 27.0 \text{ cm}^3 = 2.00 \text{ g/cm}^3$ (not 2.000)	3 sig figs (limited by volume measurement)

Key Terms

Term	Definition
Mass	The amount of matter in an object; measured in grams or kilograms
Volume	The amount of space an object occupies; measured in mL, L, or cm^3
Density	Mass per unit volume ($D = m/V$); intensive property
Accuracy	How close a measurement is to the true or accepted value
Precision	How close repeated measurements are to each other
Systematic error	Consistent error in one direction; caused by equipment or method
Random error	Unpredictable variation in either direction; caused by technique or environment
Scientific notation	Expressing numbers as $a \times 10^n$ where $1 \leq a < 10$
Significant figures	The digits in a measurement that indicate its precision
Meniscus	The curved surface of a liquid in a graduated cylinder; read at the bottom
Displacement method	Method for finding volume of irregular solids by measuring water level change

Review Questions

1. Explain the difference between mass, volume, and density of a solid. For each property, state the correct SI unit and one piece of laboratory equipment used to measure it.
2. A student is investigating the density of an irregular solid. Describe the step-by-step procedure for this investigation as it would appear in the Method section of a scientific report. Your answer must include how to measure mass and how to measure volume.
3. The following data was collected for a metal cube: mass = 89.6 g, length = 2.0 cm, width = 2.0 cm, height = 2.0 cm. Calculate the density of the cube. Show your formula, substitution, and final answer with correct units and significant figures.
4. Explain the difference between accuracy and precision. Use a dartboard analogy in your answer. Then describe a situation in which measurements could be precise but not accurate.
5. A student measures the mass of a 10.00 g standard mass five times and obtains the following readings: 9.95 g, 9.96 g, 9.94 g, 9.96 g, 9.95 g. Is this set of measurements accurate? Is it precise? Explain your reasoning using the definitions of accuracy and precision.
6. Identify four possible sources of error in a density experiment. For each source, state the type of error (systematic or random) and propose one specific improvement to enhance reliability.
7. Convert the following numbers into scientific notation:
 - (a) 0.000450
 - (b) 6,020,000,000
 - (c) 0.00000000167
8. Determine the number of significant figures in each of the following measurements:
 - (a) 0.00340 g
 - (b) 1500 mL
 - (c) 1500. mL
 - (d) 1.002×10^3 cm

9. Perform the following calculations and report the answer with the correct number of significant figures:

(a) $2.50 \text{ cm} \times 3.0 \text{ cm}$

(b) $12.11 \text{ g} + 0.3 \text{ g}$

(c) $5.00 \text{ g} \div 2.00 \text{ cm}^3$

(d) $45.6 \text{ mL} - 2.1 \text{ mL}$

10. Application question: A student calculates the density of an unknown solid as 2.85 g/cm^3 . The accepted value from a reference table is 2.70 g/cm^3 . Calculate the accuracy of the student's result (the difference from the accepted value). Then suggest two possible sources of error that could explain the discrepancy.